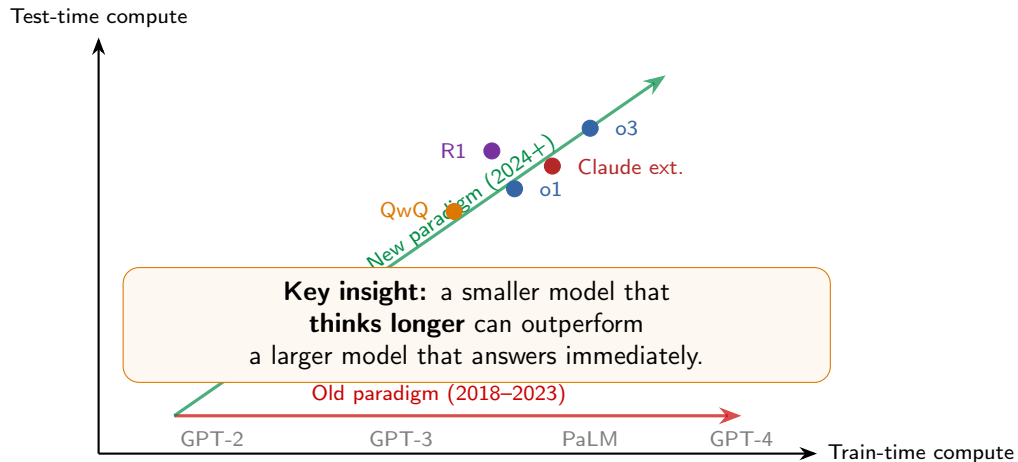


Reasoning & Test-Time Compute

Chain-of-Thought · Search · Verification · Scaling Inference

The two scaling axes



Chain-of-Thought prompting

Standard prompting

Q: Roger has 5 balls.
He buys 2 cans of 3.
How many balls?



A: 11

Chain-of-Thought prompting

Q: Roger has 5 balls.
He buys 2 cans of 3.
How many balls?



Step 1: 2 cans $\times$ 3 balls = 6 balls bought
Step 2: 5 original + 6 new = **11 balls**

Few-shot CoT (Wei et al., NeurIPS 2022):
provide exemplars with reasoning steps

Zero-shot CoT (Kojima et al., 2022): just add "*Let's think step by step*"

Emergent ability: benefits appear only at $\sim 100\text{B}+$ parameters

Why Chain-of-Thought works

Decomposition

Hard problem →
sequence of
easy sub-problems

Each step is within
the model's ability

Extra compute

More output tokens
= more FLOPs
allocated per problem

Harder problems get
more computation

System 1 vs 2

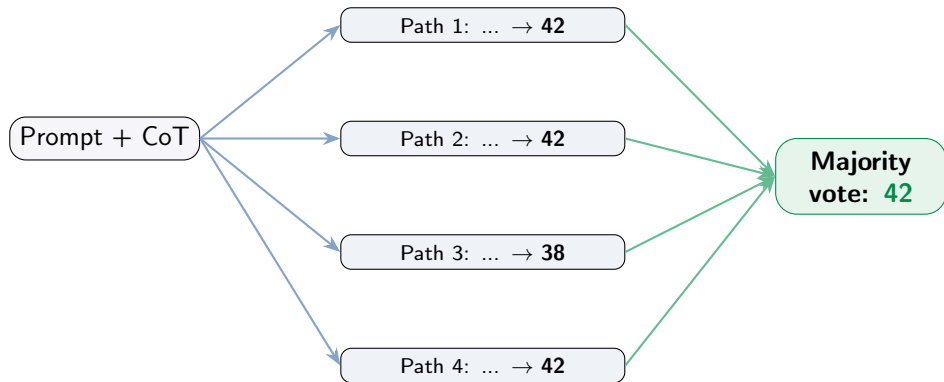
Direct answer = fast,
intuitive (System 1)

Step-by-step = slow,
deliberate (System 2)

CoT is the **simplest form** of test-time compute scaling:
generate more tokens $\Rightarrow$ allocate more computation $\Rightarrow$ better answers.

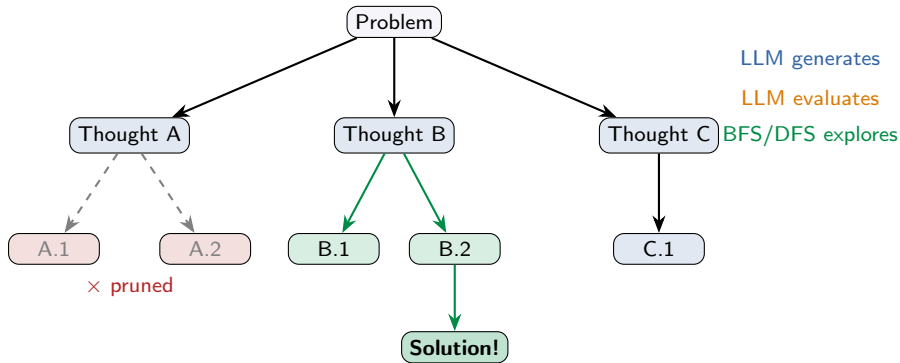
Caveat: some researchers argue CoT reflects structured inductive bias from training data, not true reasoning.

Self-Consistency



Wang et al., ICLR 2023 · **No training needed** — purely a decoding strategy
GSM8K +17.9% SVAMP +11% AQuA +12.2% StrategyQA +6.4%

Tree of Thoughts



Yao et al., NeurIPS 2023 · **Game
of 24: CoT 4% → ToT 74%**

Bridges LLM generation with classical AI search (branching + backtracking)

Process vs. Outcome Reward Models

Outcome RM (ORM)



No signal for intermediate steps (sparse) Reward: ✓ or ×

Process RM (PRM)



✓ ✓ × —

Dense, step-level feedback

Lightman et al., ICLR 2024
“Let’s Verify Step by Step”

PRM **significantly outperforms**
ORM on MATH dataset

PRM800K: 800K step-level
human labels released

PRMs enable **fine-grained search**:
score partial solutions to guide
beam search / MCTS at inference

Search at inference time

Best-of-N

Generate N candidates
Score each with
a verifier (ORM/PRM)
Return the best one

Simple but effective
Cost: $O(N)$ samples

Beam search

Expand promising
partial solutions
PRM scores each step
Prune low-scoring
branches

Finer-grained than
Best-of-N

MCTS

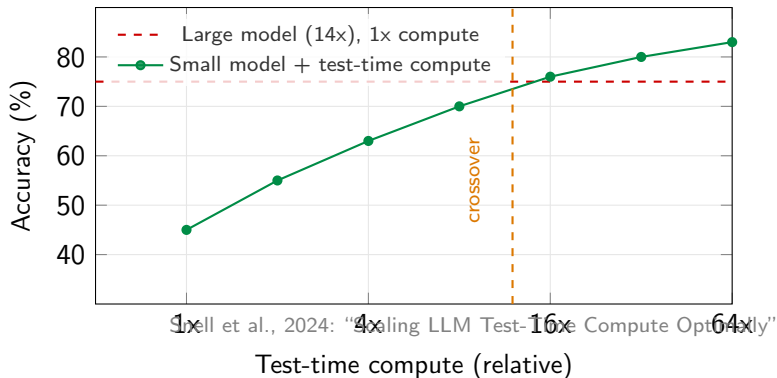
Monte Carlo Tree Search
over reasoning steps
Selection $\rightarrow$ expansion
 $\rightarrow$ rollout $\rightarrow$ backprop

7B + MCTS: **83.4%**
math (beats 72B)

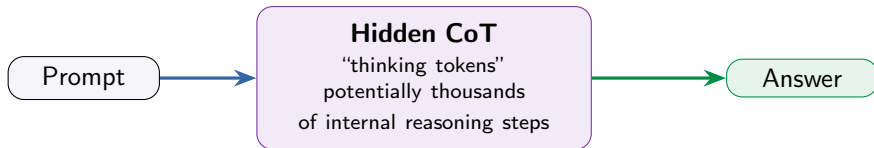
All methods share one principle: **trade
more compute for better answers.**

The key enabler is a **verifier** (PRM or self-evaluation) to distinguish good from bad reasoning.

Test-time compute scaling laws



OpenAI o1: reasoning via reinforcement learning



Trained with **large-scale RL** to produce internal reasoning chains.

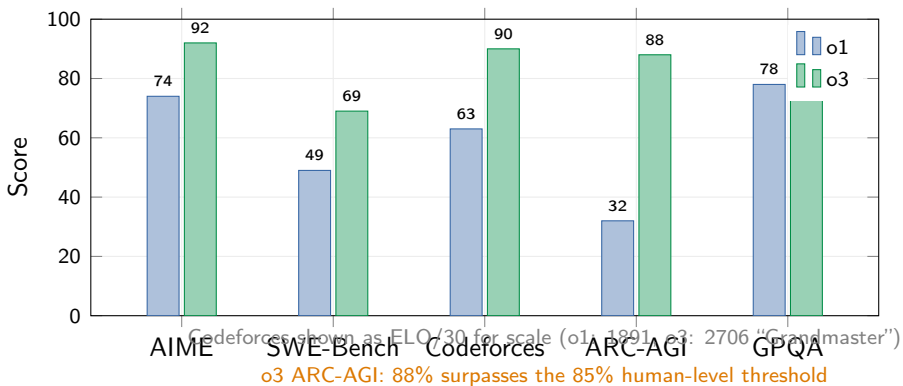
Through RL, the model learns to: refine strategies, recognize mistakes,
~~break down hard steps, and try alternative approaches~~

MATH: 94.8% **AIME 2024:** 83%
(consensus@64) **GSM8K:** ~ceiling

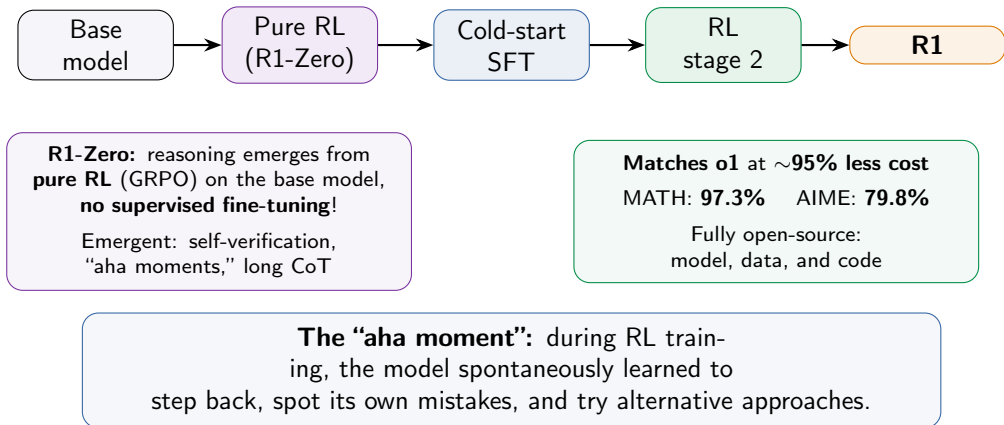
Hidden CoT is not shown to users — only

Other reasoning models: **QwQ** (Qwen, Nov 2024) — open-weight 32B reasoning model; **Claude 3.5 Sonnet extended thinking** (Anthropic, 2025) — user-controllable thinking budget.

OpenAI o3: scaling inference further

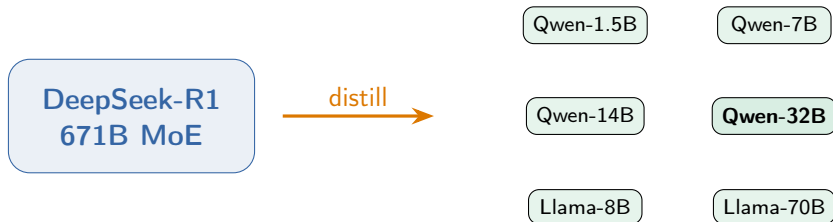


DeepSeek-R1: open-source reasoning



DeepSeek-AI, "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via RL," January 2025

Distillation of reasoning



Key finding: distilling reasoning from a large model into small models works **better** than applying RL directly to small models.

R1-Distill-Qwen-32B outperforms **o1-mini** across benchmarks.
Reasoning capability is transferable — you don't need to run RL on every model.

Budget forcing & adaptive compute

Truncation

Model exceeds token budget
⇒ forcefully end thinking
Prevents overthinking on
easy questions

Extension

Model tries to stop early
⇒ append “**Wait**” token
Forces double-checking,
often fixes errors!

s1 (Muennighoff et al., Jan 2025):
trained on just **1,000 questions**

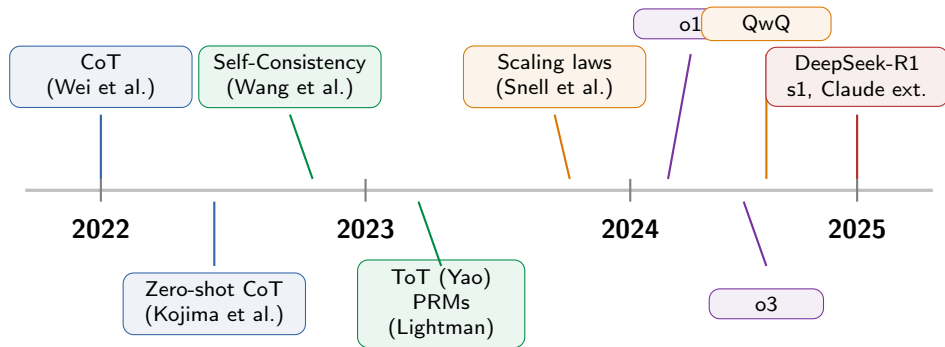
Beats **o1-preview** by up to **27%** on
competition math (MATH, AIME)

The problem: reasoning models **over-**
think easy queries and **underthink** hard ones.

Adaptive Budget Forcing: monitor confidence/entropy
in real time, allocate compute dynamically.

Claude 3.7: user-controllable “thinking to-
ken budget” · o1: multiple compute modes

The reasoning landscape



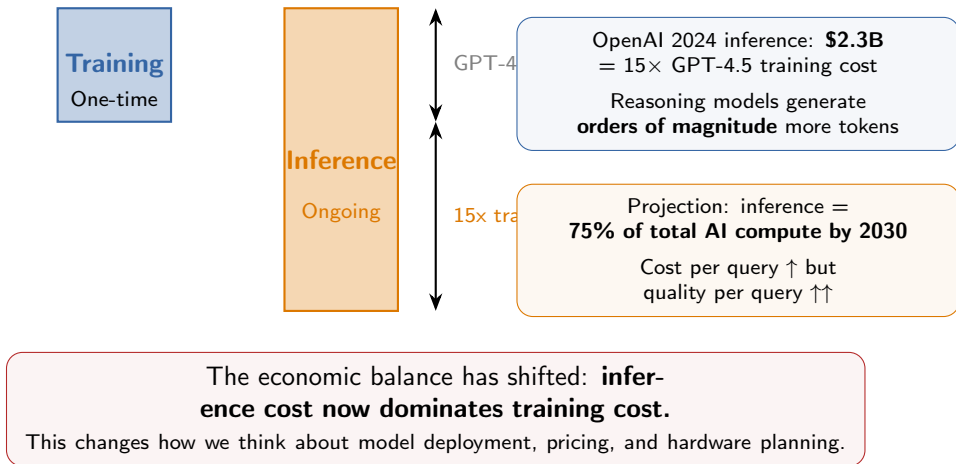
From “prompt engineering” (2022) to “inference-time scaling” (2024+) — in just 3 years.

Key reasoning benchmarks

Benchmark	What it tests	Scale	Frontier SOTA
GSM8K	Grade-school math	1K test	Saturated (>95%)
MATH	Competition math	5K test	R1: 97.3%
AIME	Hard competition (15 Q/yr)	15/year	o3: 91.6%
ARC-AGI	Fluid intelligence	~400 tasks	o3: 88%
Codeforces	Competitive programming	ELO rating	o3: 2706 (GM)
SWE-Bench	Real GitHub issues	2.3K tasks	o3: 69.1%
GPQA	PhD-level science	198 Q	o3: 87.7%

Benchmark saturation is a growing problem: GSM8K went from “hard” (2021) to “solved” (2024). New benchmarks (ARC-AGI-2, FrontierMath) keep raising the bar.

The economics of thinking



Practical guide

Simple factual Q?

→ Standard LLM
(no extra reasoning)

Multi-step reasoning?

→ CoT + self-consistency
(cheap, effective)

Math / code contest?

→ Reasoning model
(o1/o3/R1) or
MCTS+PRM

Latency-sensitive?

→ Budget forcing
(truncate/extend
adaptively)

Open-source needed?

→ DeepSeek-R1, QwQ,
or distilled variants

Want best quality?

→ Best-of-N + PRM
(scale compute
at inference)

Rule of thumb: start with CoT/self-consistency (free, universal).
Upgrade to reasoning models or search only when the task demands it.

Open questions

Efficiency

Can we get reasoning quality
without the overhead of
thousands of hidden tokens?

Allocation

How to optimally decide
how much to think
for each query?

Scaling

Will reasoning ability
keep improving with more
test-time compute, or plateau?

Generality

Is RL-discovered reasoning
truly general, or specialized
to math/code domains?

Further reading

Chain-of-Thought & Search

- Wei et al. (2022), “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”
- Lightman et al. (2024), “Let’s Verify Step by Step” (Process Reward Models)

Reasoning Models

- OpenAI (2024), “Learning to Reason with LLMs” (o1 blog post)
- DeepSeek-AI (2025), “DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via RL”
- OpenAI (2024), “o1: Reflect Deeply on the Boundaries of the Unknown”

Budget Forcing & Distillation

- Muennighoff et al. (2025), “s1: Simple Test-Time Scaling” (budget forcing)
- Yao et al. (2023), “Tree of Thoughts: Deliberate Problem Solving with LLMs”

Questions?

Next: Long Context & Efficient Attention